RESTOCK PRIORITY SCORING MODEL

Mathematical Framework & Methodology
Approval Document

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1. Executive Summary

This document presents the mathematical framework and methodology used by the Restock Priority Scoring Model, an automated system designed to determine the optimal restocking order for inventory items (SKUs) in our e-commerce platform.

The model assigns each SKU a Priority Score between 0.00 and 1.00, where a higher score indicates greater urgency for restocking. The score is derived from 8 core inventory variables, modulated by 24 external market and operational factors that dynamically adjust the relative importance (weights) of each variable.

A machine learning ensemble (XGBoost, LightGBM, and a Neural Network) is trained on this scoring framework to generalize patterns and improve prediction accuracy. The ensemble is validated against strict acceptance thresholds ($R^2 > 0.90$, MAE < 0.05) to ensure reliability.

This document is submitted for review and approval by the Financial Controller to confirm that the equations, variables, and methods are financially sound and appropriate for production use in inventory management decisions.

2. Purpose & Scope

2.1 Purpose

To automate and optimize the restocking decision process by providing a data-driven, transparent priority ranking for all inventory items. This replaces subjective manual ordering decisions with a mathematically rigorous framework.

2.2 Scope

- Covers all SKUs managed through the e-commerce platform inventory system.
- Incorporates internal inventory metrics and external market conditions.
- Produces a daily ranked restock order with actionable priority tiers.
- Integrates with the existing backend inventory management system.

2.3 Key Assumptions

- Historical sales and inventory data is accurate and up to date.
- External factor data (market indices, supplier metrics) is refreshed periodically.
- All monetary values are in a consistent base currency.
- Supplier lead times follow approximately normal distributions.

3. Master Priority Score Formula

The Priority Score (PS) for each SKU is computed as a weighted linear combination of eight normalized inventory variables:

$$PS = w_1 \cdot D + w_2 \cdot (1-S) + w_3 \cdot L + w_4 \cdot P + w_5 \cdot SC + w_6 \cdot (1-HC) + w_7 \cdot SR + w_8 \cdot SE$$

Where:

Symbol	Description
PS	Priority Score, a value in [0.00, 1.00]. Higher = more urgent.
w_1 to w_8	Dynamic weights computed from 24 external factors. Always sum to 1.00.
D	Normalized Demand Rate — how quickly the product is selling.
S	Normalized Stock Level — current inventory position. Inverted (1−S) because more stock = less urgency.
L	Normalized Lead Time — supplier delivery time. Longer = more urgent.
P	Normalized Profit Margin — product profitability. Higher margin = prioritize restocking.
SC	Normalized Stockout Cost — financial loss if product goes out of stock.
HC	Normalized Holding Cost — cost of storing inventory. Inverted (1−HC) because high storage cost = less urgency to add more.
SR	Normalized Supplier Reliability failure rate. Less reliable supplier = order sooner.
SE	Normalized Seasonality Index — seasonal demand factor. Peak season = stock up.

Constraint: $w_1 + w_2 + w_3 + w_4 + w_5 + w_6 + w_7 + w_8 = 1.00$ (weights always sum to unity)

Output Range: $0.00 \leq PS \leq 1.00$ (clipped to this range)

4. Primary Variables (8 Core Metrics)

Each of the 8 primary variables captures a distinct aspect of inventory management. Below are the detailed formulas and financial rationale for each.

4.1 Demand Rate (D)

Business Meaning: Measures how fast a product is selling. High demand products need priority restocking.

$$D = \text{Total Units Sold} / \text{Number of Days}$$

Optionally enhanced with a Weighted Moving Average (WMA) to give more importance to recent sales trends:

$$WMA = \Sigma(weight_i \times sales_i) / \Sigma(weights)$$

Financial Rationale: Revenue is directly tied to sales velocity. Products with higher demand rates generate more revenue per day, and running out of these products causes proportionally larger revenue losses.

4.2 Stock Level (S)

Business Meaning: Measures how many days of inventory remain at the current consumption rate.

$$\text{Days of Supply} = \text{Units on Hand} / \text{Demand Rate}$$

$$\text{Reorder Point} = (\text{Demand Rate} \times \text{Lead Time}) + \text{Safety Stock}$$

Note: This variable is INVERTED in the master formula (1−S) because higher stock levels mean LESS urgency to restock.

Financial Rationale: Maintaining adequate stock prevents lost sales and backorder costs, while excessive stock ties up working capital. The reorder point formula ensures orders are placed before stock is depleted, accounting for supplier lead times.

4.3 Lead Time (L)

Business Meaning: The time (in days) between placing an order and receiving it from the supplier.

$$L_avg = \text{mean(historical lead times)}$$

$$L_adjusted = L_avg + (z \times \sigma_L)$$

Where $z = 1.65$ (95% service level confidence) and σ_L is the standard deviation of historical lead times.

Financial Rationale: Unpredictable or long lead times increase the risk of stockouts. The adjusted lead time with z-score buffer provides a 95% confidence level that orders will arrive before stock is depleted, reducing emergency procurement costs.

4.4 Profit Margin (P)

Business Meaning: The profitability of each unit sold.

Contribution Margin = (Selling Price − Variable Cost) / Selling Price × 100%

Gross Margin = (Selling Price − COGS) / Selling Price × 100%

Financial Rationale: Higher margin products contribute more to bottom-line profitability per unit sold. Prioritizing restocking of high-margin items maximizes gross profit and return on inventory investment.

4.5 Stockout Cost (SC)

Business Meaning: The total financial impact when a product goes out of stock.

SC = (Price − Cost) × Q_lost + Backorder_Cost × Q_backordered + Goodwill_Loss

Component	Formula	Description
Lost Sales	(Price − Cost) × Q_lost	Profit lost from sales that could not be fulfilled
Backorder Cost	Backorder_Cost × Q_backordered	Extra cost to expedite delayed orders
Goodwill Loss	Estimated amount	Reputational damage and customer churn cost

Financial Rationale: Stockout costs are often the largest hidden cost in inventory management. This formula captures direct lost profit, operational backorder expenses, and the harder-to-quantify customer goodwill loss that impacts long-term revenue.

4.6 Holding Cost (HC)

Business Meaning: The annual cost of keeping one unit in inventory.

HC per unit = Unit Cost × Holding Rate

HC total = Average Inventory × Unit Cost × Holding Rate

Default holding rate: 25% of unit cost per year (industry standard includes warehousing, insurance, obsolescence, and opportunity cost of capital).

Note: This variable is INVERTED in the master formula (1−HC) because products with high holding costs are expensive to store, so adding more inventory is less desirable.

Financial Rationale: Holding costs represent tied-up working capital. The 25% rate is a widely accepted industry benchmark that accounts for storage, insurance, depreciation, and the opportunity cost of capital that could be deployed elsewhere.

4.7 Supplier Reliability (SR)

Business Meaning: A composite score measuring how dependable the supplier is.

SR_score = 0.4 × OTD + 0.4 × Fill Rate + 0.2 × Quality Rate

Component	Formula	Weight
On-Time Delivery (OTD)	On-Time Deliveries / Total Orders	40%
Fill Rate (FR)	Units Delivered / Units Ordered	40%
Quality Rate (QR)	Units passing inspection / Total units	20%

$$SR_failure = 1 - SR_score$$

The model uses SR_failure (unreliability) as the input — less reliable suppliers increase restock urgency because we need to order earlier to compensate for delivery uncertainty.

Financial Rationale: Unreliable suppliers increase the effective lead time and the probability of stockouts. The weighted composite balances delivery timeliness, order completeness, and product quality — all of which directly affect cost of goods and customer satisfaction.

4.8 Seasonality Index (SE)

Business Meaning: Measures how demand for a product varies by season or time period.

$$SE_t = \text{Average Demand in Period } t \,/\, \text{Overall Average Demand}$$

SE > 1.0 indicates a peak demand period (restock urgently). SE < 1.0 indicates an off-peak period (less urgency). SE = 1.0 means average demand.

Financial Rationale: Seasonal demand patterns directly impact revenue. Failing to stock adequately before peak seasons (holidays, back-to-school, etc.) results in significant lost sales. Conversely, overstocking for off-peak periods ties up capital unnecessarily.

5. External Market Factors (24 Factors)

The 24 external factors capture real-world market, financial, and operational conditions that influence how important each primary variable should be at any given time. They are organized into 8 groups, one per weight.

Group 1 — Demand Factors (→ w_1)

Code	Factor Name	Formula	Business Meaning
MV	Market Volatility	std(prices) / mean(prices)	Price instability signals uncertain demand
CCR	Customer Churn Rate	Churned / Total Customers	Customer attrition rate
DFE	Demand Forecast Error	|Actual − Forecast| / Actual	Forecasting accuracy

Group 2 — Stock Level Factors (→ w_2)

Code	Factor Name	Formula	Business Meaning
WU	Warehouse Utilization	Used Capacity / Total Capacity	How full the warehouse is
CFP	Cash Flow Pressure	1 − min(Assets/Liabilities, 1)	Financial liquidity pressure
ITR	Inventory Turnover Ratio	COGS / Average Inventory	How fast inventory cycles

Group 3 — Lead Time Factors (→ w_3)

Code	Factor Name	Formula	Business Meaning
SCDI	Supply Chain Disruption	1 − (LPI Score / 5)	Logistics disruption risk
IDR	Import Dependency Ratio	Imported / Total Units	Reliance on imports
GRI	Geopolitical Risk Index	min(GPR Index / 200, 1)	Political risk to supply chain

Group 4 — Profit Margin Factors (→ w_4)

Code	Factor Name	Formula	Business Meaning
CPP	Competitive Price Pressure	max(0, (Competitor−Ours)/Ours)	Pricing competitiveness
PED	Price Elasticity of Demand	|%ΔDemand / %ΔPrice|	Price sensitivity of demand
IBG	Industry Benchmark Gap	max(0, (Industry−Ours)/Industry)	Margin gap vs industry

Group 5 — Stockout Cost Factors (→ w_5)

Code	Factor Name	Formula	Business Meaning
MCI	Market Competition Index	1 − (HHI / 10000)	Competitive intensity
CSC	Customer Switching Cost	1 − Switching Ease Rate	Cost for customers to switch
BLI	Brand Loyalty Index	((NPS+100)/200 + Repeat Rate) / 2	Customer loyalty strength

Group 6 — Holding Cost Factors (→ w_6)

Code	Factor Name	Formula	Business Meaning
IR	Interest Rate Factor	min(Central Rate / 0.20, 1)	Cost of capital
SCI	Storage Cost Index	min(Current / Benchmark, 1)	Relative storage cost
PPR	Product Perishability Rate	Expired / Total Held	Spoilage/obsolescence rate

Group 7 — Supplier Factors (→ w_7)

Code	Factor Name	Formula	Business Meaning
SCR	Supplier Concentration	Top Supplier / Total Units	Single-supplier dependency
GSCV	Global Supply Volatility	1 − (LPI / 5)	Global logistics stability
SFH	Supplier Financial Health	min(Credit Score / 850, 1)	Supplier creditworthiness

Group 8 — Seasonality Factors (→ w_8)

Code	Factor Name	Formula	Business Meaning
SDCV	Seasonal Demand CV	std(demand) / mean(demand)	Demand variability across seasons
ISI	Industry Seasonality Index	min(Swing% / 100, 1)	Industry-wide seasonal effect
CEI	Climate Event Impact	Event-Driven Revenue %	Weather/climate impact on demand

6. Dynamic Weight Computation

The 24 external factors are combined into 8 raw weights using the following formulas. Each formula groups 3 related factors to determine the importance of the corresponding primary variable.

Weight	Raw Formula	Rationale
w_1 (Demand)	(MV + CCR + DFE) / 3	Equal average of demand-related factors
w_2 (Stock)	0.4×WU + 0.3×CFP + 0.3×(1/max(ITR, 0.001))	Weighted blend; low turnover increases urgency
w_3 (Lead Time)	(SCDI + IDR + GRI) / 3	Equal average of supply chain risk factors
w_4 (Profit)	0.4×CPP + 0.3×PED + 0.3×IBG	Competitive pressure weighted highest
w_5 (Stockout)	(MCI + (1−CSC) + (1−BLI)) / 3	Inverted loyalty: easy-to-lose customers increase cost
w_6 (Holding)	0.3×IR + 0.3×SCI + 0.4×PPR	Perishability weighted highest
w_7 (Supplier)	(SCR + GSCV + (1−SFH)) / 3	Inverted health: weak suppliers increase risk
w_8 (Season)	0.4×SDCV + 0.3×ISI + 0.3×CEI	Demand variability weighted highest

Normalization Step

$$w_i\ (final) = w_i\ (raw) / \Sigma(w_1_raw + w_2_raw + \ldots + w_8_raw)$$

This ensures all weights always sum to exactly 1.00, maintaining the priority score within the [0, 1] range regardless of market conditions.

Fallback Weights

When external factor data is unavailable, the system uses pre-defined fallback weights based on standard inventory management priorities:

Weight	Variable	Fallback Value	Justification
w_1	Demand Rate	0.25 (25%)	Demand is the primary driver of restocking needs
w_2	Stock Level	0.20 (20%)	Current stock is the second most critical factor
w_3	Lead Time	0.15 (15%)	Supply chain timing affects order planning
w_4	Profit Margin	0.10 (10%)	Profitability guides capital allocation
w_5	Stockout Cost	0.10 (10%)	Cost of stockout affects financial exposure
w_6	Holding Cost	0.08 (8%)	Storage costs affect working capital
w_7	Supplier Reliability	0.07 (7%)	Supplier risk is a secondary factor
w_8	Seasonality	0.05 (5%)	Seasonal effects are periodic adjustments

Total: 0.25 + 0.20 + 0.15 + 0.10 + 0.10 + 0.08 + 0.07 + 0.05 = 1.00 ✓

7. Normalization Method

All primary variables are normalized to the range [0, 1] using Min-Max scaling to ensure comparability across different units and scales (e.g., days vs. dollars vs. percentages).

Standard Normalization

$$X_normalized = (X - X_min) / (X_max - X_min)$$

Applied to: D, L, P, SC, SR, SE

Inverted Normalization

$$X_normalized = 1 - (X - X_min) / (X_max - X_min)$$

Applied to: S (Stock Level), HC (Holding Cost)

Inversion is applied because for these variables, a lower raw value indicates higher restock urgency (less stock = more urgent; lower holding cost = cheaper to store more).

8. Machine Learning Ensemble Method

While the deterministic formula above provides a transparent baseline, we employ a machine learning ensemble to capture non-linear patterns and interactions between variables that the linear formula cannot express.

8.1 Three Base Models

Model	Type	Purpose
XGBoost	Gradient-boosted decision trees	Captures feature interactions and non-linearities
LightGBM	Leaf-wise gradient boosting	Fast training with high accuracy on tabular data
MLP Neural Network	2-layer fully connected network (128→64→1)	Learns complex non-linear mappings

8.2 Meta-Learner (Stacking)

The three base model predictions are combined using Ridge Regression (a regularized linear model) as a meta-learner:

$$\text{Final Score} = \text{Ridge}(\text{XGBoost_pred}, \text{LightGBM_pred}, \text{MLP_pred})$$

The meta-learner is trained on Out-of-Fold (OOF) predictions using 5-fold Time-Series Cross-Validation to prevent data leakage and overfitting.

8.3 Why an Ensemble?

- Reduces prediction variance — individual model errors cancel out.
- Captures both linear relationships (Ridge) and non-linear patterns (trees, neural network).
- More robust to noisy or incomplete input data.

- Validated against the deterministic formula to ensure consistency.

9. Priority Tier Classification

The continuous Priority Score is mapped to four actionable business tiers:

Tier	Score Range	Required Action	Response Window
CRITICAL	0.75 – 1.00	Restock immediately	Within 24 hours
HIGH	0.50 – 0.74	Place priority purchase order	Within 3 days
MEDIUM	0.25 – 0.49	Schedule routine restock	Within 1 week
LOW	0.00 – 0.24	Monitor only, no action	Monthly review

These thresholds can be adjusted based on business requirements and risk tolerance. The current values represent a balanced trade-off between service level targets and inventory carrying costs.

10. Model Validation & Acceptance Criteria

The model must pass all of the following acceptance thresholds before deployment. These criteria are evaluated on a held-out test set (15% of data, never seen during training).

Metric	Threshold	Description	Industry Standard
Mean Absolute Error (MAE)	< 0.05	Average prediction error must be under 5%	Standard regression benchmark
Root Mean Square Error (RMSE)	< 0.08	Penalizes large errors more heavily	Standard regression benchmark
R-Squared (R^2)	> 0.90	Model explains >90% of score variance	High bar for financial models
Mean Absolute % Error (MAPE)	< 7%	Percentage-based error metric	Acceptable for demand planning
Precision @ Top 10%	> 0.85	85%+ of predicted critical items are truly critical	Ensures restocking accuracy
Recall @ Top 10%	> 0.90	90%+ of truly critical items are detected	Minimizes missed stockouts

Ongoing Monitoring

The system includes automated drift detection:

- RMSE Drift Alert: Triggered if RMSE increases by more than 0.03 from baseline.
- R^2 Drift Alert: Triggered if R^2 drops by more than 0.05 from baseline.
- Population Stability Index (PSI): Monitors input feature distribution changes.

11. Approval Sign-Off

By signing below, the approver confirms that the mathematical equations, variables, weighting methodology, machine learning approach, and priority tier classifications described in this document are:

- Mathematically sound and correctly implemented.
- Financially appropriate for inventory management decision-making.
- Consistent with standard inventory management and supply chain best practices.
- Suitable for production deployment in the e-commerce platform.

Prepared By:	**Reviewed By:**
Name: _____	Name: _____
Title: _____	Title: Financial Controller
Date: _____	Date: _____
Signature: _____	Signature: _____

☐ **APPROVED**

☐ **APPROVED WITH MODIFICATIONS (attach comments)**

☐ **NOT APPROVED (attach reasons)**

Comments / Modifications:

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